Trade Policy Uncertainty and the Chilean Peso

A portfolio model and joint empirical evidence

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Theory, exchange-rate dynamics, and a common-sample mechanism test

Research question

Does U.S. trade-policy uncertainty reprice the peso through risk compensation?

- The model links current depreciation to greater terminal currency risk when investors bear positive peso exposure.
- The empirical test aligns the price, terminal variance, and survey target on the same 258 news months.

Empirical preview

The positive price estimate builds after the news month.
The joint evidence leaves the risk-compensation mechanism unresolved.

Baseline: January 2003–September 2024. Common three-month origins: January 2003–June 2024.

Assets and terminal currency value

$$Q_0 = \text{USD per CLP}, \qquad E_0 = 1/Q_0 = \text{CLP per USD}.$$

Asset	Current dollar cost	Terminal dollar payoff
Dollar asset	1	R^*
Peso asset	Q_0	RQ_1

$$Q_1 = \mu(u) + \sigma(u)\varepsilon, \qquad \Pr(\varepsilon = \pm 1) = \tfrac{1}{2}, \quad 0 < \sigma < \mu.$$

Pure risk increases σ while holding μ fixed.

$R, R^* > 0$ are gross interest factors. A falling Q_0 means peso depreciation.

Net peso exposure and risk tolerance

$$U_i(W) = -e^{-\gamma_i W}, \qquad \gamma_i > 0.$$

$$\boxed{K = B + \sum_i h_i, \qquad T = \sum_i \frac{1}{\gamma_i}}$$

- $B > 0$: peso-asset supply held by the pricing sector.
- Rh_i: background peso income or obligations.
- K: net peso exposure after market clearing.
- T: aggregate absolute risk tolerance.

The sign of exposure determines the currency's risk premium.

Borrowing, lending and short positions are unrestricted. Portfolio approach: Gabaix and Maggiori (2015).

Market clearing prices the currency

$$\boxed{Q_0 = \frac{R}{R^*}\left[\mu - \sigma \tanh\left(\frac{KR\sigma}{T}\right)\right]}$$

Expected excess dollar payoff per asset

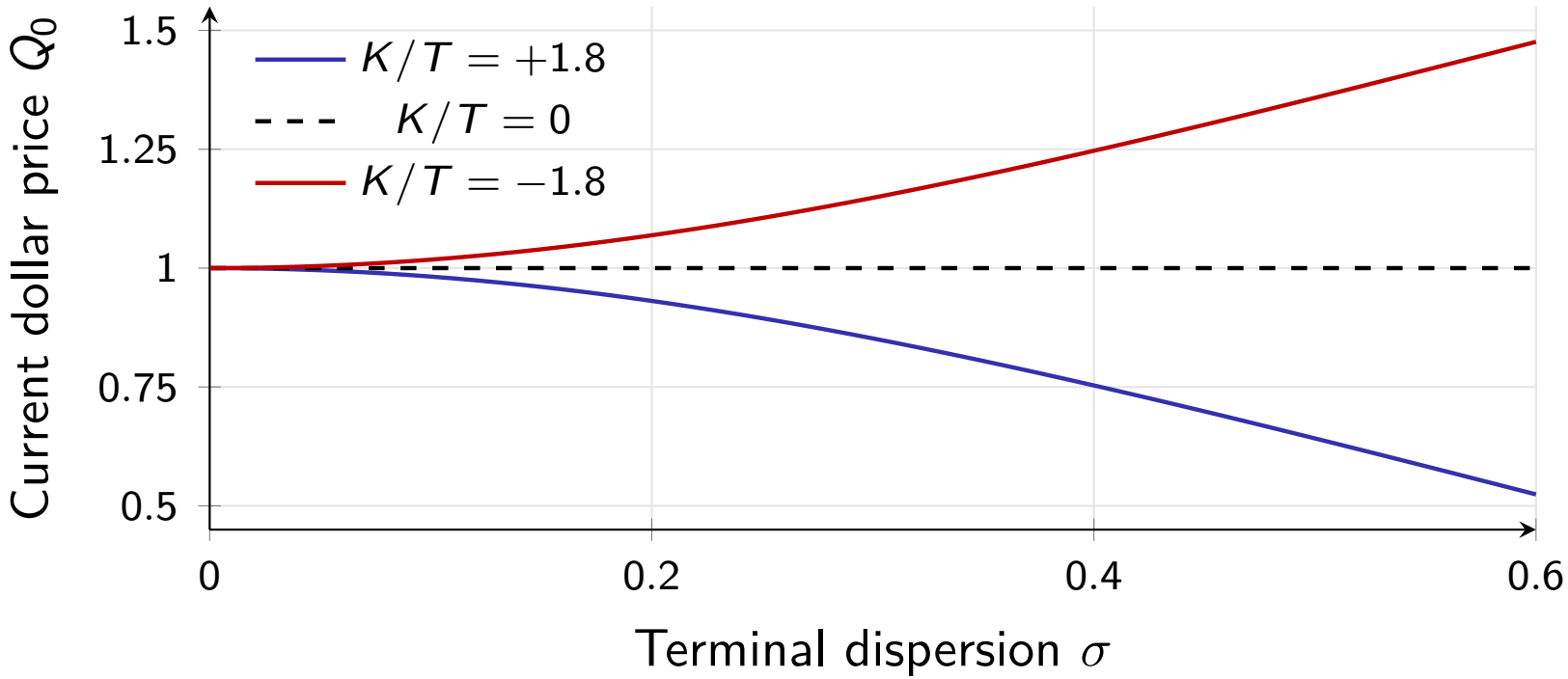
$$\rho = R\mu - R^* Q_0 = R\sigma \tanh\left(\frac{KR\sigma}{T}\right).$$

Expected simple dollar excess return

$$\mathbb{E}_0\left[\frac{RQ_1}{Q_0}\right] - R^* = \frac{\rho}{Q_0}.$$

The exact equilibrium lies strictly between the discounted low- and high-state payoffs. Proof in backup.

The price effect depends on net exposure



With $K > 0$, more risk lowers the peso's dollar price.

Illustrative equilibria with $\mu = R = R^* = 1$. The horizontal axis varies dispersion, not time.

Expected-value news can reinforce or offset repricing

Let $z = KR\sigma/T$ and $G(z) = \tanh z + z\,\mathrm{sech}^2 z$.

$$\frac{dQ_0}{du} = \frac{R}{R^*}\left[\underbrace{\mu'(u)}_{\text{expected-value news}} - \underbrace{\sigma'(u)G(z)}_{\text{risk compensation}}\right].$$

- Adverse expected-value news reinforces depreciation.
- Favorable expected-value news can offset the risk adjustment.

The empirical test needs risk and expectations alongside prices.

Rates, net exposure and risk tolerance are fixed in this comparative static.

Theory and the empirical comparison

Model implication	Empirical object
Current peso repricing	News-month CLP/USD depreciation
Greater terminal currency risk	Conditional variance of dollar value at $t+3$
Expected terminal value	EEE forecast tilts for the same $t+3$ target
Higher expected compensation	Matched simple dollar excess returns

The main table aligns the first three objects on 258 origins.

Survey tilts compare forecasts with reference-price anchors. They are proxies for expected-value adjustment. Net exposure K remains unmeasured. Matched returns remain a data extension.

Data and common support

- **TPU**: Caldara et al.'s U.S. newspaper index.
- **Peso**: BCCh reference rate, with trading and effective dates retained.
- **Expectations**: EEE two-month question in $t+1$, targeting $t+3$.
- **Market risk**: daily spot returns, LSEG `CLP3MO=`, and Bloomberg tenor/RR quotes.

Estimation support	Months	Origins
TPU first stage	261	Jan. 2003–Sep. 2024
Main joint comparison	**258**	Jan. 2003–Jun. 2024

Sources: Caldara et al. (2020), Banco Central de Chile, LSEG and Bloomberg. The common comparison retains the original innovation scale.

Empirical specification

A standardized TPU innovation

$$U_t = \log(1 + \mathrm{TPU}_t) = a + \Pi' W_t + \xi_t, \qquad \eta_t = \widehat{\xi}_t / \widehat{s}.$$

Cumulative price response

$$100(\log E_{t+h} - \log E_{t-1}) = \alpha_h + \beta_h \eta_t + \Gamma_h' W_t + e_{t,h}.$$

Terminal conditional variance

$$Y_{t,3} = 100(Q_{t+3}/Q_{t-1} - 1), \qquad \log \mathrm{Var}(Y_{t,3} \mid \eta_t, W_t) = c + \gamma_3 \eta_t + \lambda' Z_t.$$

W_t: 16 lagged currency, risk and macro variables.
Z_t: lagged log realized variance and lagged log VIX.

The terminal mean is linear in η_t and W_t. Estimates are conditional innovation responses. Inference retains serial dependence and the estimated innovation scale. Jordà (2005).

Joint evidence on 258 common news months

One TPU standard deviation, one four-outcome confidence family

Response and economic unit	Estimate	Simultaneous 95% CI
News-month depreciation, log pp	−0.0124	[−0.4949, 0.4701]
Terminal variance at $t+3$, %	+2.08	[−27.91, 44.56]
Survey remaining tilt, USD-value pp	−0.1857	[−0.3973, 0.0260]
Survey total tilt, USD-value pp	−0.1066	[−0.4997, 0.2866]

Current depreciation and terminal variance remain imprecise.

999 paired 12-month block draws. Variance: log coefficient 0.0206 with interval $[-0.3273, 0.3685]$, transformed by $100(e^{\gamma}-1)$. Survey values share the pre-news dollar-value denominator. Source: archived joint258 results.

The joint pricing criterion

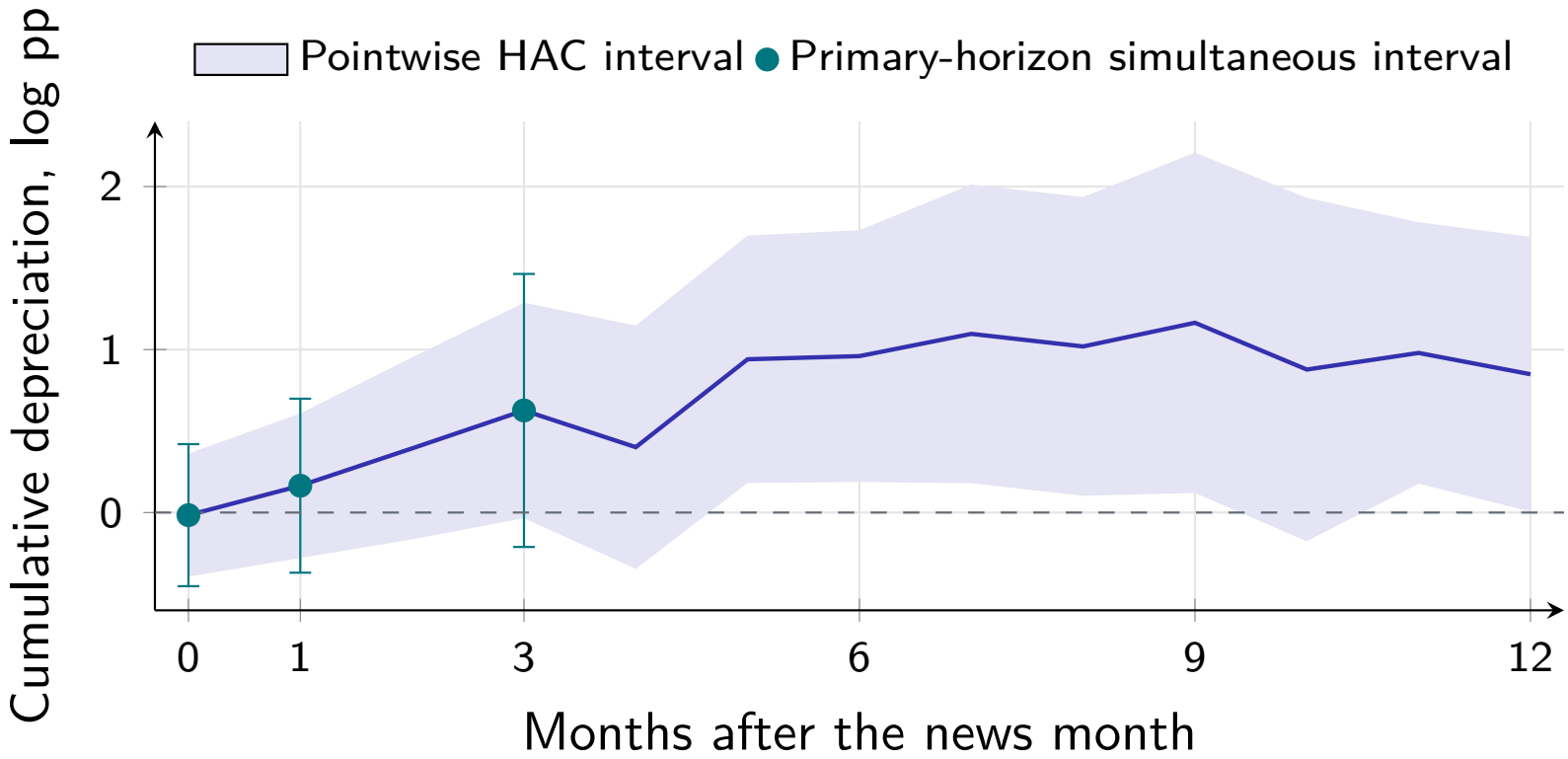
Required restriction	Evidence
Terminal variance lower bound > 0	No
Current depreciation lower bound > 0	No
Both survey intervals inside the chosen bound	Yes
All restrictions jointly	**No**

The archived bounds ± 0.5, ± 1 and ± 2 pp give the same joint decision.

The positive price and risk restrictions are not established.

The bounds concern survey-response slopes. Survey point medians do not identify the model's arithmetic conditional mean. Decisions use unrounded simultaneous limits.

The price point estimate builds after the news month



At $h = 3$: **0.626 log pp**, simultaneous interval $[-0.211,\ 1.464]$.

Cumulative window: end of $t - 1$ through $t + 3$. Separate price family over $h = 0, 1, 3$; 9,999 draws, blocks of 12. The $h = 3$ row uses the same 258 origins as the main table.

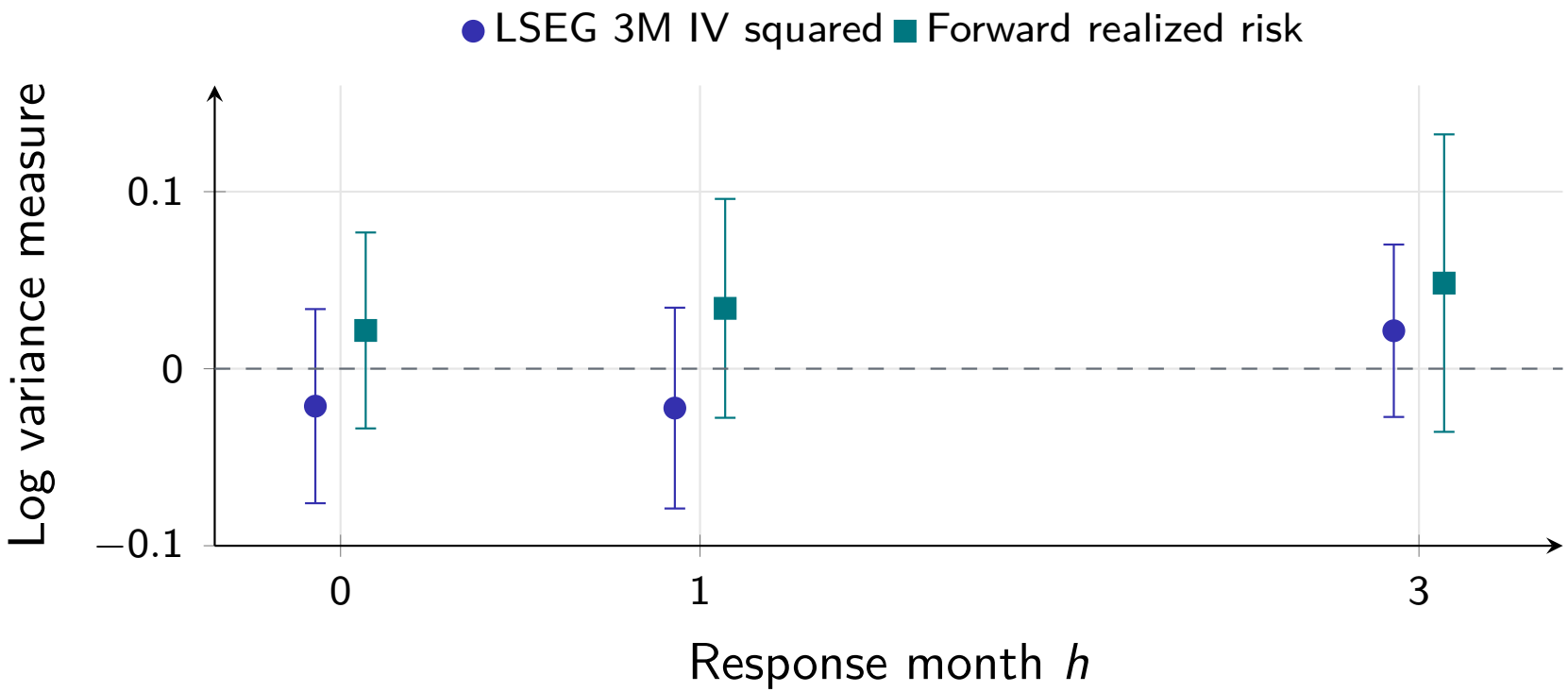
The price estimate is sensitive to 2018–2019

Three-month cumulative price specification	Log pp
Baseline	0.626
Six dynamic lags	0.588
Monthly-average reference rate	0.554
Excluding windows overlapping 2018–2019	0.203

- The price point estimate is similar across several routine specifications.
- Effective-rate and peer-currency comparisons provide weaker evidence.

Episode exclusions measure concentration. Each alternative reestimates its innovation. Full sensitivity estimates are in the accompanying result archive.

Market risk provides a supplementary comparison



Neither measure shows a precise positive response.

Six-margin simultaneous intervals, Bartlett 12. Matched $N = 252, 251, 249$. IV is a rolling 3M quote at $t + h$; realized risk covers the next three calendar months. Source: LSEG forward-risk comparison.

Additional option tenors and quote construction

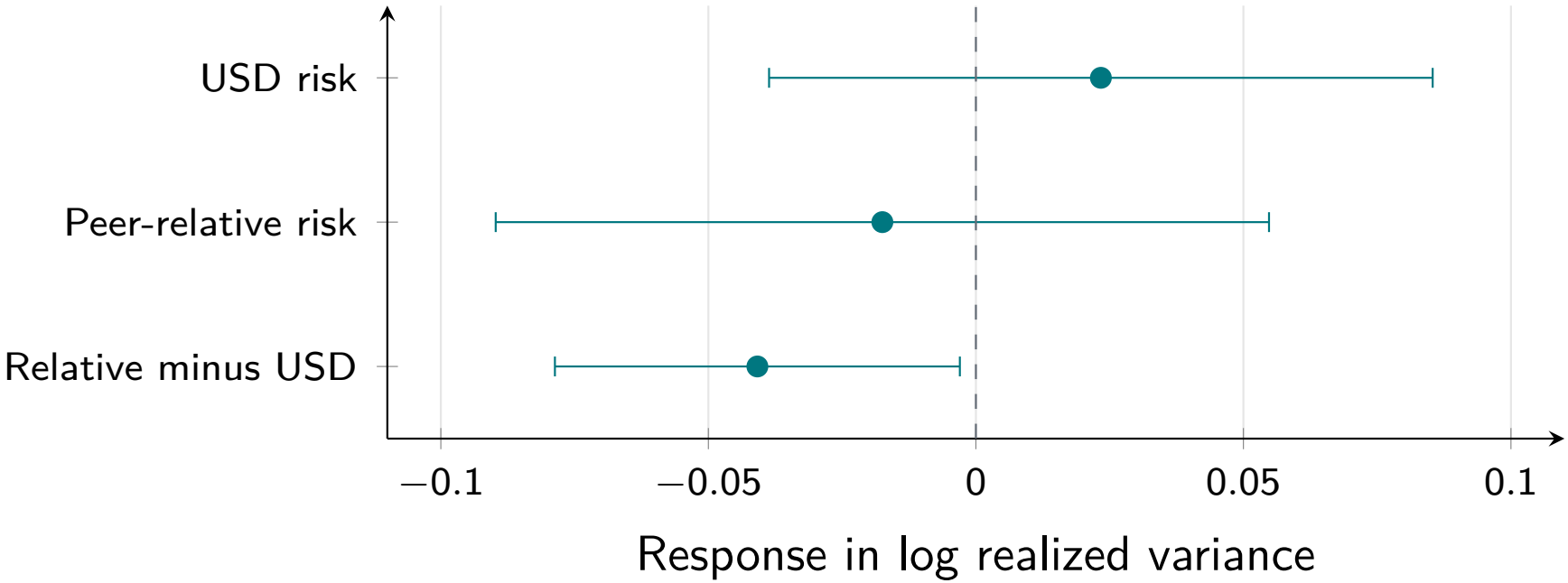
- Bloomberg 1M, 2M, 6M and 12M volatility levels show no stable positive response.
- Signed 3M risk-reversal intervals also contain zero.

Aggregation changes the measured $h = 3$ response

Monthly mean minus month-end	Estimate	Simultaneous 95% CI
6M log squared midquote	-0.0164	$[-0.0326, -0.0002]$
12M log squared midquote	-0.0163	$[-0.0287, -0.0039]$

Aggregation compares $\log(\text{mean}_d\, IV_d^2)$ with $2 \log IV_{\text{EOM}}$ on the same sample. 36-margin family, Bartlett 12. Supplier labels and signed RR units retained. Sources: Bloomberg curve, RR and aggregation results.

Denomination changes the path-risk response



Relative-to-USD risk ratio: **about 4% lower** per TPU standard deviation.

252 common origins and one six-margin family. Relative-minus-USD coefficient −0.0408 [−0.0787, −0.0030]. Peer basket: AUD, CAD, NZD, BRL, MXN. Absolute-risk responses remain imprecise.

Conclusion

- The bilateral price point response builds after the news month.
- On 258 common origins, the evidence does not establish current depreciation and higher terminal variance together.
- Broader market-risk measurements reinforce the distinction between terminal payoff risk and realized path fluctuations.

Next empirical priority

Separate policy news about expected fundamentals from changes in dispersion. Matched cash-flow excess returns would directly test compensation.

Selected references

- Caldara, D., M. Iacoviello, P. Molligo, A. Prestipino and A. Raffo (2020). The economic effects of trade policy uncertainty. *Journal of Monetary Economics* 109, 38–59.
- Gabaix, X. and M. Maggiori (2015). International liquidity and exchange rate dynamics. *Quarterly Journal of Economics* 130, 1369–1420.
- Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review* 95, 161–182.
- Ludvigson, S., S. Ma and S. Ng (2021). Uncertainty and business cycles: exogenous impulse or endogenous response? *AEJ: Macroeconomics* 13, 369–410.

Backup: optimization and the unique equilibrium

Let $a = R\mu - R^* Q_0$ and $d = R\sigma$.

$$CE_i = C_i + n_i a - \frac{1}{\gamma_i} \log\cosh[\gamma_i d(n_i + h_i)].$$

$$CE_i' = a - d\tanh[\gamma_i d(n_i + h_i)], \qquad CE_i'' < 0.$$

The finite asset demand is

$$n_i^d = \frac{1}{\gamma_i d}\operatorname{artanh}(a/d) - h_i, \qquad |a| < d.$$

Summing and imposing $\sum_i n_i = B$ gives

$$\frac{T}{d}\operatorname{artanh}(a/d) = K, \qquad a = d\tanh(Kd/T).$$

The equilibrium lies inside $R(\mu - \sigma)/R^* < Q_0 < R(\mu + \sigma)/R^*$. Aggregate demand decreases strictly across this interval.

Backup: risk sharing and a local variance approximation

Equilibrium risk sharing

$$n_i^{eq} = \frac{K}{\gamma_i T} - h_i, \qquad n_i^{eq} + h_i = \frac{K}{\gamma_i T}.$$

Fixed supply and exposure permit repricing with unchanged positions.

Small exposure: $|KR\sigma/T| \ll 1$

$$\rho \simeq \frac{KR^2}{T}\sigma^2, \qquad Q_0 \simeq \frac{R}{R^*}\mu - \frac{KR^2}{TR^*}\sigma^2.$$

$$\frac{\partial Q_0}{\partial K} = -\frac{R^2\sigma^2}{R^* T}\operatorname{sech}^2 z, \qquad \frac{\partial Q_0}{\partial T} = \frac{KR^2\sigma^2}{R^* T^2}\operatorname{sech}^2 z.$$

The local approximation exposes how variance, exposure and risk tolerance enter the price adjustment.

Backup: definitions of the common-target survey tilts

$\widetilde{Q}_{t+1,3}$: reciprocal of the median positive EEE response in $t+1$, targeting $t+3$.

Q_{t+1}^{pre}: latest reference price effective before the survey month's first day.

$$f_t^{post} = 100\frac{\widetilde{Q}_{t+1,3} - Q_{t+1}^{pre}}{Q_{t-1}}, \qquad f_t^{total} = 100\left(\frac{\widetilde{Q}_{t+1,3}}{Q_{t-1}} - 1\right).$$

- Both quantities share the news-month origin and $t+3$ terminal target.
- Both use the pre-news dollar price Q_{t-1} as denominator.
- They compare a point forecast with price anchors.

These are not consecutive-survey forecast revisions. In the ± 0.5 grid, the total-tilt lower limit is -0.4997409; the interval-containment decision uses the unrounded value.

Backup: controls and simultaneous inference

Sixteen nonconstant controls

- Three lags each of log TPU, peso returns and log realized variance.
- Lagged log VIX and the reference-rate level.
- Broad-dollar and copper changes, Fed funds, Chilean policy rate and IMACEC growth.

Inference families

- Main comparison: four margins, 999 paired replications with 12-month blocks.
- Original price profile: $h = 0, 1, 3$, 9,999 replications at block lengths 12 and 24 months.
- Market-risk supplements: their archived, separate simultaneous families.

Blocks resample already aligned tuples and reestimate the TPU stage and standard deviation. Most extensions are exploratory. Montiel Olea and Plagborg-Møller (2021) motivate lag-augmentation sensitivity.

Backup: survey forecasts contain predictive information

EEE two-month question, expressed in dollar-value changes
ΔMAE = survey minus benchmark.

Benchmark	N	Δ MAE	Joint 95% CI
Pre-survey flat price	219	-0.327	$[-0.614, -0.040]$
Recursive three-lag price rule	156	-0.462	$[-0.837, -0.087]$

- Survey forecasts also have lower squared error in these comparisons.
- The survey's advantage over the flat-price benchmark has no precise association with TPU.

MAE in normalized USD-value percentage points. Two-loss paired HAC families, Bartlett 12. Training for the recursive rule uses already completed targets. Sources: EEE flat-skill and three-lag benchmark results.

Backup: directional amplitude and IV forecasting

Terminal directional amplitude

$$C = (L - \bar{a}_- D) - (G - \bar{a}_+(1 - D)), \quad L = \max(-Y, 0), \quad G = \max(Y, 0).$$

At $t + 3$, the centered downside-minus-upside slope is

$$\boxed{0.590 \text{ pp} \qquad [0.084,\ 1.097]}.$$

Historical risk forecasting

- Adding IV reduces absolute log-risk error from 0.4256 to 0.3900 on 189 origins.
- Adding lagged TPU to the IV rule gives no precise loss improvement on 134 common origins.

The amplitude contrast measures directional composition. The main conditional-variance result is separate. Sources: partial-moment side contrast and recursive IV-risk forecast results.